BrineWatch

An autonomous sentinel for the hidden cost of fresh water

Low-cost robotic mapping and live digital twins of desalination brine plumes

Prototypes for Humanity 2026
Project proposal

Applicant & team lead: Andrea Bedei — PhD student, University of Bologna (MSc Automotive Engineering for Intelligent Mobility, 2025) · **Academic supervisors:** Giovanni Pau and Roberto Girau, University of Bologna · **Industrial partner:** MBS (marine robotics)

In one sentence: a $5,000 open-source underwater robot that finds, maps and watches the hypersaline plumes desalination plants discharge into the sea — turning an invisible externality into a live, verifiable digital twin.

1 The problem: fresh water's invisible waste stream

More than 300 million people drink water taken from the sea. Desalination is the lifeline of water-scarce regions — and it is growing fast, in the Gulf and now across the Mediterranean. But every litre of fresh water leaves behind ≈1.5 litres of **brine**: hypersaline, warmer, laced with antiscalant and chlorine residues, returned to the sea through submarine outfalls at ≈142 million m^3 *per day* worldwide; four Gulf states alone produce over half of it [1], into the worst possible basin — shallow, semi-enclosed, already the saltiest open sea on Earth.

The physics makes the harm invisible. Brine is *denser* than seawater: leaving the diffuser it collapses onto the seabed and creeps downslope as a gravity current [2], forming hypersaline bottom layers that suffocate benthic life — seagrass meadows, shellfish, the base of coastal fisheries. It also feeds a vicious cycle: a saltier intake makes every litre more energy-hungry, which produces more brine. And none of this can be seen from above: satellites and surface buoys observe the surface, while the damage happens at the bottom.

2 Why today's monitoring fails

Discharge permits are granted on **CFD predictions made once, on paper**, and rarely verified against reality. In operation, compliance rests on two or three **fixed buoys** — single points, usually at the wrong depth for a bottom-hugging plume — and on **vessel campaigns every 3–6 months**, in which a crew lowers a probe over a grid of stations: tens of thousands of euros per survey for a few dozen point measurements, with months of darkness in between. Permits define a **mixing zone** (typically: salinity within ∼5% of ambient at 100–300 m from the diffuser), yet no instrument in routine use can actually verify it in space and time. The result: regulators enforce limits they cannot measure, operators get no feedback to correct themselves, and *nobody has ever seen the real 3-D shape of a plume evolving over time.*

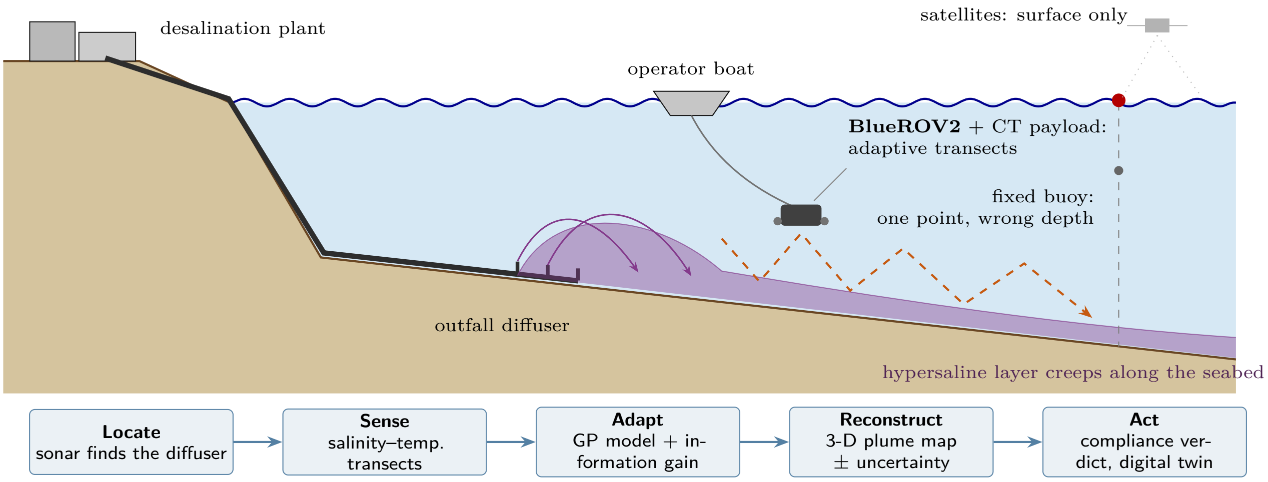

Figure 1: **The BrineWatch mission.** Dense brine sinks and creeps along the seabed, where satellites and fixed buoys cannot see it. A $5k open-source ROV crosses the plume boundary with adaptive transects and turns every mission into an updated 3-D map, a compliance verdict and a living digital twin of the discharge zone.

3 Our approach: a robot that follows the salt

BrineWatch turns a $5k-class open-source ROV — a **BlueROV2 Heavy** with low-light HD camera, side-scan sonar (Cerulean Omniscan SS450) and IMU — plus a ≈€300 **conductivity–temperature payload** (I^{2}C on BlueOS) into an autonomous compliance instrument (Fig. 1):

1. **Locate** — the sonar finds the outfall pipe and diffuser on the seabed, anchoring the survey frame.
2. **Sense** — continuous salinity–temperature–depth sampling along baseline transects.
3. **Adapt** — an online Gaussian-Process field model picks the next waypoints by expected information gain [3]:

the robot concentrates effort where it matters — the plume boundary — *following the salt* instead of mowing a blind grid.
4. **Reconstruct** — the posterior becomes a 3-D isohaline map with explicit uncertainty.
5. **Act** — the map is checked automatically against permit thresholds; every mission updates a **living digital twin** of the discharge zone: trends, seasonal drift, alerts, and a feedback loop the plant can use to modulate dilution or discharge timing.

The planner is vehicle-agnostic (waypoint interface, MAVLink-ready): it ports to any ROV/AUV, and tomorrow to residency stations for fully unattended operation.

4 Measurable advantages

	Today (vessel + buoys)	**BrineWatch**
Cost per survey	€20–50k (vessel, crew, winch)	≈€500 all-in per mission
Frequency	every 3–6 months	weekly or on demand
Data	a few dozen point casts	thousands of samples; full 3-D field + uncertainty
Output	static report, months later	same-day compliance verdict; live digital twin
Barrier to entry	survey vessel and specialist crew	$5k open-source ROV — any municipality or lab

Adaptive sampling is itself a quantified gain: informative path planning reports 2–3× efficiency over fixed grids in field monitoring [3]; our simulation study benchmarks exactly this for benthic brine plumes — same battery, lawnmower vs. adaptive — and reports reconstruction error side by side.

5 Validation plan: honest and time-boxed

Phase 1 — by submission (simulation-complete). Full-mission demonstration in **HoloOcean 2.x** [4] (Unreal Engine 5.3): official BlueROV2 Heavy model, ray-cast side-scan and forward-looking sonar, configurable currents. The plume is an analytic negatively-buoyant jet + seabed gravity current (Roberts-type parameterization [2]) advected by an idealized tide; a virtual CT sensor samples the field at the vehicle pose with realistic noise. Deliverables: mission video, lawnmower-vs-adaptive benchmark at equal battery, dashboard with automatic compliance verdict. We deliberately declare the analytic plume: in deployment, BrineWatch is precisely the instrument that *validates CFD models against reality.*

Phase 2 — with award support. Integration of the CT payload on our physical BlueROV2 (≈€300 sensor), tank and harbour trials, then a pilot survey at a partner desalination outfall with a utility or environmental regulator — the natural hosts are in the Gulf itself.

6 Real-world impact

Water security. Desalination capacity will keep growing with climate stress — and brine with it. Making discharge verifiable builds the social licence for the technology hundreds of millions of people drink from; the intake-salinity feedback also protects plant efficiency. **Ecosystems and livelihoods.** Continuous bottom-layer monitoring protects seagrass and benthic habitats — and the artisanal fisheries above them. **Democratized enforcement.** At two orders of magnitude below vessel-survey cost, any coastal city, regulator or university lab can hold an outfall accountable — in the Gulf, the Mediterranean, or anywhere the sea is asked to pay for fresh water. **Beyond brine.** The same platform audits sewage outfalls, desalination intakes and port infrastructure: one sentinel for the submerged interface between cities and the sea. *(SDG 6 · 14 · 9)*

[1] E. Jones *et al.*, “The state of desalination and brine production: a global outlook,” *Sci. Total Environ.* 657, 2019. [2] P. J. W. Roberts, “Near-field flow dynamics of concentrate discharges and diffuser design,” in *Intakes and Outfalls for Seawater RO Desalination*, Springer, 2015. [3] G. Hitz *et al.*, “Adaptive continuous-space informative path planning for online environmental monitoring,” *J. Field Robotics* 34(8), 2017. [4] E. Potokar *et al.*, “HoloOcean: an underwater robotics simulator,” *IEEE ICRA*, 2022; “A preview of HoloOcean 2.0,” arXiv:2510.06160, 2025.